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Inferences in MLR:

Model

$$y = X\beta + \varepsilon$$

$$A^{-1} = (X^T X)^{-1}$$

The matrix has same order as A (i.e $(K+1) \times (K+1)$) and is written as

$$\begin{bmatrix} C_{00} & C_{01} & \cdots & C_{0K} \\ C_{10} & C_{11} & \cdots & C_{1K} \\ \vdots & \vdots & \cdots & \vdots \\ C_{K0} & C_{K1} & \cdots & C_{KK} \end{bmatrix} \quad -①$$

σ^2 (variance) unknown, we can estimate it by using sample variance.

$$S^2 = \sigma^2 = \frac{SSE}{N-K-1} \quad OR \quad \frac{\Sigma (y_i - \hat{y}_i)^2}{N-K-1}$$

$$b = A^{-1} g$$

$$b = \begin{bmatrix} b_0 \\ \vdots \\ b_K \end{bmatrix}$$

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b_j is normally distributed with

- mean = β_j
- variance = $\sigma^2 C_{jj}$

The diagonal of matrix ① is C_{jj}

Hypothesis Testing:

Hypothesis:

$H_0 : \beta_j = \beta_{j0}$

$H_1 : \beta_j \neq \beta_{j0}$

$H_1 : \beta_j > \beta_{j0}$

$H_1 : \beta_j < \beta_{j0}$

Test statistics:

$$t_0 = \frac{b_j - \beta_{j0}}{S\sqrt{C_{jj}}}$$

b_j : Estimated coefficient
β_{j0} : Hypothesized value of coefficient
C_{jj} : j^{th} diagonal of matrix

Critical Region:

① $t \geq t_{\alpha/2}$ or $t \leq -t_{\alpha/2}$

② $t > t_\alpha$

③ $t < -t_\alpha$

$V = N - K - 1$

N : number of obs.
K : number of regressors.

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Variable Screening Using T-test

Hypothesis:

$H_0 : \beta_j = 0$

$H_1 : \beta_j \neq 0$

Test Stats:

$$t_0 = \frac{b_j - \beta_{j0}}{s\sqrt{C_{jj}}}$$

Critical Region:

$t \geq t_{\alpha/2}$

or

$t \leq -t_{\alpha/2}$

if we Reject H_0, it means j^{th} regressor is significant and we can't drop it.

- Helps identity which predictors significantly contribute to explaining y.

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EXAMPLE:

Test the hypothesis that β2 = −2.5 at the 0.05 level of significance against the alternative that β2 > −2.5

Table 12.3: Data for Example 12.4

y (% survival)	x_1 (weight %)	x_2 (weight %)	x_3 (weight %)
25.5	1.74	5.30	10.80
31.2	6.32	5.42	9.40
25.9	6.22	8.41	7.20
38.4	10.52	4.63	8.50
18.4	1.19	11.60	9.40
26.7	1.22	5.85	9.90
26.4	4.10	6.62	8.00
25.9	6.32	8.72	9.10
32.0	4.08	4.42	8.70
25.2	4.15	7.60	9.20
39.7	10.15	4.83	9.40
35.7	1.72	3.12	7.60
26.5	1.70	5.30	8.20

Hypothesis:

$H_0 : \beta_2 = -2.5$

$H_1 : \beta_2 > -2.5$

Test stats:

$$t = \frac{b_2 - \beta_{20}}{s\sqrt{c_{22}}}$$

$$= \frac{-1.8616 + 2.5}{2.073\sqrt{0.0166}}$$

$$= 2.390$$

Critical region:

$\alpha = 0.05$

$v = n - k - 1$

$= 13 - 3 - 1$

$= 9$

$t_{(9)}(0.05) = 1.833$

$2.390 > 1.833$

Reject H_0.

EXAMPLE:

1. Determine a multiple linear regression model

2. Perform individual variable screening using T-test

Site	x_1	x_2	x_3	x_4	x_5	y
1	15.57	2463	472.92	18.0	4.45	566.52
2	44.02	2048	1339.75	9.5	6.92	696.82
3	20.42	3940	620.25	12.8	4.28	1033.15
4	18.74	6505	568.33	36.7	3.90	1003.62
5	49.20	5723	1497.60	35.7	5.50	1611.37
6	44.92	11,520	1365.83	24.0	4.60	1613.27
7	55.48	5779	1687.00	43.3	5.62	1854.17
8	59.28	5969	1639.92	46.7	5.15	2160.55
9	94.39	8461	2872.33	78.7	6.18	2305.58
10	128.02	20,106	3655.08	180.5	6.15	3503.93
11	96.00	13,313	2912.00	60.9	5.88	3571.59
12	131.42	10,771	3921.00	103.7	4.88	3741.40
13	127.21	15,543	3865.67	126.8	5.50	4026.52
14	252.90	36,194	7684.10	157.7	7.00	10,343.81
15	409.20	34,703	12,446.33	169.4	10.75	11,732.17
16	463.70	39,204	14,098.40	331.4	7.05	15,414.94
17	510.22	86,533	15,524.00	371.6	6.35	18,854.45

1) Multiple linear Regression:

$$\hat{y} = \beta_0 + \beta_1 x_1 + \beta_2 x_2 + \beta_3 x_3 + \beta_4 x_4 + \beta_5 x_5 \quad -(1)$$

as $\hat{\beta} = (X^T X)(X^T Y)$

* β_0 : 1710.7679
* β_1 : -9.6249
* β_2 : 0.0563
* β_3 : 1.3772
* β_4 : -3.9881
* β_5 : -358.0028

Put in eq (1)

MLR Model:

$$\hat{y} = 1710.7679 - 9.6249x_1 + 0.0563x_2 + 1.3772x_3 - 3.9881x_4 - 358.0028x_5.$$

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2) Variable screening:

with $\alpha = 0.05$

For β_1:

$H_0 : \beta_1 = 0$

$H_1 : \beta_1 \neq 0$

$\Rightarrow$ Test statistics:

$$t_0 = \frac{\hat{\beta}_1}{SE(\hat{\beta}_1) \leftarrow S\sqrt{C_{11}}} = \frac{-9.6249}{96.210}$$

$$\boxed{= -0.100}$$

$\Rightarrow$ Critical Region. $v = N-K-1 = 15$

$t_{0.025}(15) \boxed{= 2.131}$

$t_0 \geq t_{\alpha/2}(v) = -0.100 \geq 2.131$ False

$t_0 \leq -t_{\alpha/2}(v) = -0.100 \leq -2.131$ False

$\Rightarrow$ Conclusion: Accept $H_0 \rightarrow$ x_1 regressor is insignificant.

For β_2:

$H_0 : \beta_2 = 0$

$H_1 : \beta_2 \neq 0$

$\Rightarrow$ Test statistics:

$$t_0 = \frac{\hat{\beta}_2}{SE(\hat{\beta}_2) \leftarrow S\sqrt{C_{22}}} = \frac{0.0563}{0.021}$$

$$\boxed{= 2.685}$$

$\Rightarrow$ Critical Region: $v = N-K-1 = 15$

$t_{0.025}(15)$ $\boxed{= 2.131}$

$$\left[\begin{array}{ll} t_0 \geq t_{\alpha/2}(v) & 2.685 \geq 2.131 \text{ (True)} \\ t_0 \leq -t_{\alpha/2}(v) & 2.685 \leq -2.131 \text{ (False)} \end{array}\right]$$

$\Rightarrow$ Conclusion: ~~Acce~~ Reject H_0

$\boxed{\text{regressor } x_2 \text{ is significant.}}$

For β_3:

$H_0 : \beta_3 = 0$

$H_1 : \beta_3 \neq 0$

$\Rightarrow$ Test statistics:

$$t_0 = \frac{\hat{\beta}_3}{S\sqrt{C_{33}}} = \frac{1.3772}{3.047}$$

$SE(\hat{\beta}_3) \leftarrow S\sqrt{C_{33}}$

$\boxed{= 0.452}$

$\Rightarrow$ Critical Region: $v = N-K-1 = 15$

$t_{0.025}(15)$ $\boxed{= 2.131}$

$$\left[\begin{array}{lll} t_0 \geq t_{\alpha/2}(v) & 0.452 \geq 2.131 & \text{False} \\ t_0 \leq -t_{\alpha/2}(v) & 0.452 \leq -2.131 & \text{False} \end{array}\right]$$

$\Rightarrow$ Conclusion: Accept H_0 $\rightarrow$ $\boxed{x_3 \text{ regressor is insignificant.}}$

For β_4:

$H_0 : \beta_4 = 0$

$H_1 : \beta_4 \neq 0$

$\Rightarrow$ Test statistics

$$t_0 = \frac{\hat{\beta}_4}{S\sqrt{C_{44}}} = \frac{-3.9881}{7.061} \quad \boxed{= -0.565}$$

$SE(\hat{\beta}_4) \leftarrow S\sqrt{C_{44}}$

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=) Critical Region: $V = N-K-1 = 15$

$t_{0.025}(15) = \boxed{2.131}$

$t_0 \geq t_{\alpha/2}(V)$ $\quad -0.565 \geq 2.131$ (False)

$t_0 \leq t_{\alpha/2}(V)$ $\quad -0.565 \leq -2.131$ (False)

=) Conclusion: H_0 Accept -> x_4 regressor is insignificant.

For β_5:

$H_0: \beta_5 = 0$

$H_1: \beta_5 \neq 0$

=) Test Statistics:

$t_0 = \dfrac{\hat{\beta}_5}{SE(\hat{\beta}_5)} = \dfrac{-358.0028}{207.106}$

$SE(\hat{\beta}_5) \leftarrow s\sqrt{C_{55}}$

$\boxed{= -1.729}$

=) Critical Region: $V = N-K-1 = 15$

$t_{0.025}(15) = \boxed{2.131}$

$t_0 \geq t_{\alpha/2}(V)$ $\quad -1.729 \geq 2.131$ False

$t_0 \leq -t_{\alpha/2}(V)$ $\quad -1.729 \leq -2.131$ False

=) Conclusion: H_0 Accept -> x_5 regressor is insignificant.

With the help of individual variable screening we can conclude that x_2 is most significant in this model.